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within Buildings.Fluid.Chillers.Examples.BaseClasses;
partial model PartialElectric
  "Base class for test model of chiller electric EIR"
package Medium1 = Buildings.Media.Water "Medium model";
package Medium2 = Buildings.Media.Water "Medium model";

parameter Modelica.Units.SI.Power P_nominal
  "Nominal compressor power (at y=1)";
parameter Modelica.Units.SI.TemperatureDifference dTEva_nominal=10
  "Temperature difference evaporator inlet-outlet";
parameter Modelica.Units.SI.TemperatureDifference dTCon_nominal=10
  "Temperature difference condenser outlet-inlet";
parameter Real COPc_nominal = 3 "Chiller COP";
parameter Modelica.Units.SI.MassFlowRate mEva_flow_nominal
  "Nominal mass flow rate at evaporator";
parameter Modelica.Units.SI.MassFlowRate mCon_flow_nominal
  "Nominal mass flow rate at condenser";

Buildings.Fluid.Sources.MassFlowSource_T sou1(
  redeclare package Medium = Medium1,
  use_T_in=true,
  m_flow=mCon_flow_nominal,
  T=298.15) "Mass flow source"
  annotation (Placement(transformation(extent={{-60,6},{-40,26}})));
Buildings.Fluid.Sources.MassFlowSource_T sou2(
  redeclare package Medium = Medium2,
  use_T_in=true,
  m_flow=mEva_flow_nominal,
  T=291.15) "Mass flow source"
  annotation (Placement(transformation(extent={{60,-6},{40,14}})));
Buildings.Fluid.Sources.Boundary_pT sin1(
  redeclare package Medium = Medium1,
  nPorts=1) "Pressure source"
  annotation (Placement(
    transformation(
      extent={{10,-10},{-10,10}},
      origin={70,40})));
Buildings.Fluid.Sources.Boundary_pT sin2(
  redeclare package Medium = Medium2,
  nPorts=1) "Pressure source"
  annotation (Placement(
    transformation(
      extent={{-10,-10},{10,10}},
      origin={-70,-20})));
Modelica.Blocks.Sources.Ramp TSet(
  duration=3600,
  startTime=3*3600,
  offset=273.15 + 10,
  height=8) "Set point for leaving chilled water temperature"
  annotation (Placement(transformation(extent={{-80,50},{-60,70}})));
Modelica.Blocks.Sources.Ramp TCon_in(
  height=10,
  offset=273.15 + 20,
  duration=3600,
  startTime=2*3600) "Condenser inlet temperature"
  annotation (Placement(transformation(extent={{-90,10},{-70,30}})));
Modelica.Blocks.Sources.Ramp TEva_in(
  offset=273.15 + 15,
  height=5,
  duration=3600,
  startTime=3600) "Evaporator inlet temperature"
  annotation (Placement(transformation(extent={{50,-40},{70,-20}})));
Modelica.Blocks.Sources.Pulse pulse(period=3600/2)
  "Pulse signal generator"
  annotation (Placement(transformation(extent={{-80,80},{-60,100}})));
Modelica.Blocks.Logical.GreaterThreshold greaterThreshold(threshold=0.5)
  "Greater threshold"
  annotation (
    Placement(transformation(extent = {{-40, 80}, {-20, 100}})));
Buildings.Fluid.FixedResistances.PressureDrop res1(

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    redeclare package Medium = Medium1,
    m_flow_nominal=mCon_flow_nominal,
    dp_nominal=6000) "Flow resistance"
    annotation (Placement(transformation(extent={{32,30},{52,50}})));
Buildings.Fluid.FixedResistances.PressureDrop res2(
    dp_nominal=6000,
    redeclare package Medium = Medium2,
    m_flow_nominal=mEva_flow_nominal) "Flow resistance"
    annotation (Placement(transformation(extent={{-20,-30},{-40,-10}})));
equation
    connect(TCon_in.y, sou1.T_in) annotation (Line(
        points={{-69,20},{-62,20}},
        color={0,0,127},
        smooth=Smooth.None));
    connect(TEva_in.y, sou2.T_in) annotation (Line(
        points={{71,-30},{80,-30},{80,8},{62,8}},
        color={0,0,127},
        smooth=Smooth.None));

    connect(greaterThreshold.u, pulse.y) annotation (Line(
        points={{-42,90},{-59,90}},
        color={0,0,127},
        smooth=Smooth.None));
    connect(res1.port_b, sin1.ports[1]) annotation (Line(
        points={{52,40},{60,40}},
        color={0,127,255},
        smooth=Smooth.None));
    connect(res2.port_b, sin2.ports[1]) annotation (Line(
        points={{-40,-20},{-60,-20}},
        color={0,127,255},
        smooth=Smooth.None));
    annotation (Diagram(coordinateSystem(extent={{-100,-100},{100,120}})), Icon(
        coordinateSystem(extent={{-100,-100},{100,120}})));
end PartialElectric;

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